



OPEN Wearable sensing for badminton stroke recognition with one-dimensional convolutional neural network

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Motivated by the need to improve the performance of badminton players, various motion monitoring systems have been developed to assist coaches in badminton technique instruction. While traditional video or optical methods are limited to fixed scenarios and require tedious postprocessing, IMUs, with their compact size and lightweight design, offer a portable solution for badminton motion monitoring. This study develops a wearable sensing network consisting of two IMUs to capture wrist motions during badminton strokes. The measurements are fed into a one-dimensional convolutional neural network (1D-CNN) to classify six stroke actions and fifteen badminton movement trajectories. The network includes two convolutional layers with 8 and 16 filters (3×1 kernel size), followed by batch normalization and ReLU activation. It also features a max-pooling layer, a fully connected layer, and a Softmax classifier. The network was trained with cross-entropy loss and the Adam optimizer (learning rate = 0.005, batch size = 64 iterating over 200 epochs). Experiments were conducted with six national-level badminton athletes to collect motion data for network training and prediction and stratified 5-fold cross-validation was employed to divide the training and testing datasets, ensuring the stability of model evaluation and the reliability of the results. Feature differences between the patterns were intuitively displayed using heatmaps and t-SNE approaches. The results show that the proposed method achieves classification accuracies of $97.16 \pm 0.6\%$ for six stroke actions and $86.07 \pm 1.1\%$ for fifteen badminton trajectories, outperforming traditional machine learning methods such as KNN (94.94% / 78.71%), SVM (94.32% / 75.06%), and DTA (92.81% / 72.93%). By combining a simple sensor configuration (two sensors) with a lightweight 1D-CNN, this study achieves accurate badminton motion monitoring, which is expected to contribute to sports training and enhance athletes' skills.

Keywords Wearable sensing, Badminton, Machine learning, Human motion recognition

Abbreviations

IMU	Inertial measurement Units
CNN	Convolutional neural network
ESP32	Espressif systems 32-bit
FC	Forehand clear
FD	Forehand drop
FS	Forehand smash
FL	Forehand lift
FNT	Forehand net tumble
FNT	Forehand cross-net shot
FSC	Forehand straight clear
FSD	Forehand straight drop shot
FSS	Forehand straight smash
FCC	Forehand cross-court clear
FCD	Forehand cross-court drop shot
FCS	Forehand cross-court smash
OCC	Overhead cross-court clear
OCD	Overhead cross-court drop shot

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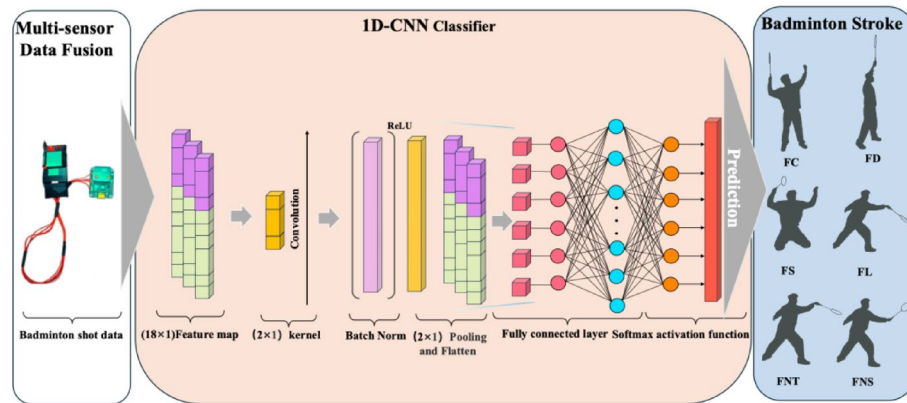


Fig. 1. Pattern recognition of different stroke actions and trajectories based on inertial gravity unit and 1D Convolution Neural Network.

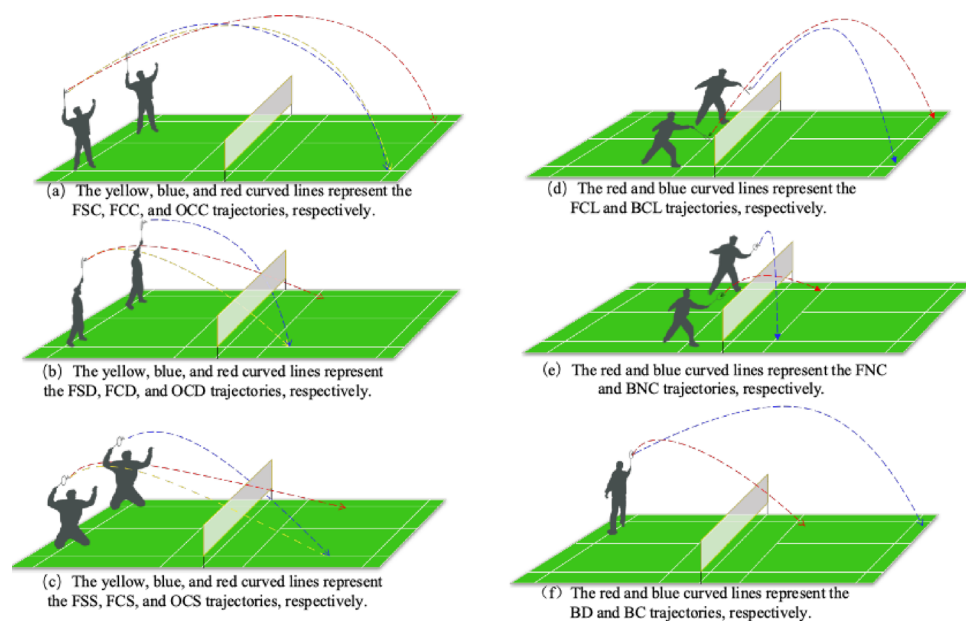


Fig. 2. (a), (b), (c), (d), (e), (f) compare fifteen different stroke actions trajectories.

OCS	Overhead cross-court smash
FCL	Forehand cross-court lift
BCL	Backhand cross-court lift
FNC	Forehand net cross shot
BNC	Backhand net cross shot
BD	Backhand drop shot
BC	Backhand clear shot

Badminton stroke is a key motion pattern that involves coordinated bone and muscle activity, enabling athletes to perform effectively in the game. It also plays a crucial role in achieving success in high-level badminton competitions¹. Although many methods have been proposed for recognizing different stroke actions, real-world training and competition environments often introduce significant disturbances. Therefore, most existing recognition techniques still rely heavily on video analysis. Using a single sensor typically limits the data to one variable, such as acceleration, making it insufficient to fully capture the biomechanical complexity of human motion^{2,3}. This study proposes a multi-sensor data acquisition and synchronization method. For experimental validation, we use a one-dimensional convolutional neural network (1D-CNN) to recognize six types of badminton stroke actions (Fig. 1) and fifteen distinct shuttle trajectories (Fig. 2). In recent years, the rapid development of Internet of Things (IoT)-based smart healthcare systems has significantly driven the growing demand for wearable devices⁴. As a key component of such systems, wearable IoT (WIoT) devices generate massive amounts of individual health data. Various types of sensors have been widely applied in human motion

monitoring, providing more accurate and reliable medical information^{5,6}. These technologies demonstrate great potential in fields such as fall detection⁷, finger motion monitoring, early diagnosis of Parkinson's disease⁸, and smart clothing⁹. In particular, they play a crucial role in the health management of the elderly population. Moreover, similar motion recognition technologies can be applied to sports, where precise action analysis can significantly improve athletic performance. Inspired by previous studies, this work further investigates the detailed recognition and analysis of badminton stroke actions and their corresponding trajectories. The accurate recognition of six fundamental badminton stroke actions—Forehand Clear, Forehand Drop, Forehand Smash, Forehand Lift, Forehand Net Tumble, and Forehand Cross-Net Shot—is essential for achieving optimal competitive performance and enhancing shot quality, particularly among lower-level players. These six stroke types serve as the foundation for generating fifteen distinct shuttlecock trajectories. Each trajectory differs in terms of angle, speed, and landing position, which are influenced by how the stroke is executed and the player's tactical intent during the match. Consequently, the ability to recognize both the executed stroke and the resulting trajectory plays a pivotal role in developing athletes' tactical awareness and playing styles¹⁰. Specifically, the recognition of stroke types and trajectories is crucial for improving the accuracy of shuttlecock landing spots after the shot and for reducing the likelihood of unforced errors. Thus, accurately predicting changes in stroke actions has a key effect on improving athletes' overall competitive abilities. However, there are significant differences in the motion biomechanics of the human body when participants perform different badminton stroke actions. Specifically, the dynamic patterns of the three actions—clear, smash, and drop—differ. Additionally, when playing badminton, athletes experience eccentric muscle contractions¹¹. Excessive eccentric contractions may lead to damage of the muscle cell membranes, triggering an inflammatory response that results in a decline in muscle function¹². By improving stroke technique, athletes can enhance the efficiency of the proximal-to-distal joint kinetic chain, thereby effectively reducing joint load and minimizing the risk of injuries caused by excessive eccentric contractions¹³. Therefore, during daily training, participants should optimize their stroke actions based on correct technical strategies to prevent injury. In swimming training, activated muscle patterns are critical for enhancing performance, and similarly, in badminton training, activating key muscle groups is also important for improving athletic performance¹⁴. Similar to monitoring in swimming^{15,16}, the real-time feedback system can help badminton athletes acquire an efficient and feasible method to correct mistakes in different stroke actions, which rely on the distinct biomechanical characteristics of various stroke actions. Recognition of badminton stroke actions has evolved through three stages: manual visual observation, wearable device data analysis, and combining wearable devices with artificial intelligence technology^{17,18}. The high cost of traditional video analysis and sensor data analysis makes it difficult to apply to the rapidly changing environment of badminton competitions and individual training. Inertial measurement units (IMUs), characterized by high portability^{19,20}, provide a wearable approach for monitoring limb joint motion to predict badminton stroke actions and stroke trajectories based on joint angle changes during training. Yu et al. proposed a simple dual-sensor setup capable of accurately identifying gait phases during unrhymic activities such as walking, stair ascending, and descending, and distinguishing these movements from standing still²¹, demonstrating that IMUs can be used for complex dynamic motion segmentation. However, during long-term use, IMUs experience data drift, leading to a decline in data accuracy²². Recently, studies have explored the combination of Kalman filtering and sensors, enhancing the stability of data during dynamic motion²³.

Recently, as the Internet of Things (IoT) has developed, the role of data integration has been increasingly recognized²⁴. Beniczky et al. have shown that the fusion of wearable physiological signals with traditional EEG data within machine learning frameworks, can significantly enhance seizure prediction accuracy²⁵. In addition, the effectiveness of swimming posture recognition²⁶, gesture recognition^{27,28}, and clinical rehabilitation has been validated through integration of machine learning techniques. Specifically, in the medical field, the use of a wearable device that combines the microcontroller and IMUs for knee joint angle recognition has reduced treatment costs for patients²⁹. The use of neural networks and deep learning is not limited to image recognition; it can also be widely applied in the field of sport^{30,31}. As of now, no study has proposed multi-sensor data acquisition and synchronization for different badminton stroke actions and stroke trajectories. Although video analysis is widely used, relevant research in this area is still scarce. Acceleration induced by changes in skeletal joint angles was detected by a single sensor for the recognition of the smash shot during the final phase of the badminton stroke¹⁰. However, human movements are accomplished through the coordination of multiple bones and skeletal muscles, and a single sensor has limitations in capturing the collaborative work of multiple joints; therefore, the coordinated operation of multiple sensors enables the capture of joint movements across multiple joints during a badminton stroke³², allowing for the detection of more subtle joint angle variations. At the same time, it is necessary to avoid the excessive computational load on the computer caused by the using too many sensors. Specifically, in a multi-IMU pattern, the spatial positions and motion trajectories of multiple joints can be effectively captured throughout dynamic movements³³. Human limb motion is generated by the contraction of skeletal muscles, which pull on tendons connected to bones, resulting in coordinated joint movement. Therefore, utilizing IMUs to monitor joint rotations during the execution of a badminton stroke provides a viable and practical method for capturing kinematic data. Based on the above, this study integrates the ESP32 microcontroller (Espressif Systems 32-bit) with IMUs (Inertial Measurement Unit) sensors to recognize different badminton stroke types using a one-dimensional convolutional neural network (1D-CNN). Considering that various badminton strategies are composed of different shot trajectories; this work further investigates shot trajectory classification based on stroke recognition. The main contributions of this study are as follows:

- 1) This work is the first to propose the use of an ESP32 microcontroller in conjunction with multi-sensor data acquisition and synchronization techniques for recognizing badminton stroke actions and different shot trajectories, thereby laying a foundation for further research into badminton tactics.

- 2) The proposed method achieved recognition accuracies of 97.16% for six badminton stroke actions and 86.07% for fifteen distinct shot trajectories, respectively.
- 3) The multi-sensor data acquisition and synchronization method addresses the issues of data discrepancies among multiple sensors and the insufficiency of single-joint motion features, thereby enabling the accurate recognition of badminton stroke actions and distinct shot trajectories.

The rest of the study is organized as follows: Section II discusses the application and potential of IMUs and the ESP32 microcontroller in badminton and other sports. Section III presents the multi-sensor data acquisition and synchronization method for badminton stroke recognition, including the strokes and shot trajectories to be recognized. Section IV introduces the hardware setup and experimental protocol. Section V validates and analyzes the stroke actions and trajectories experimentally using the proposed method. Section VI discusses the results. Section VII provides the conclusion.

Related work

In this section, we conduct in-depth exploration of the status quo of badminton action recognition and related applications of the Internet of Things in sports. The research progress related to badminton posture recognition can be seen from Table 1.

State of machine learning in sports

Juyi Lin et al.³⁴ used IMUs in combination with a voiceprint extraction module to accurately classify six hitting actions from ten athletes. The median accuracy rates after 10-fold cross-validation for models using four methods—Random Forest (RF), SMO with a polynomial kernel, SMO with an RBF kernel, and Naive Bayes—were 78%, 84%, 80%, and 77%, respectively. In comparison, our study employs two IMUs along with a deep learning approach to recognize fifteen different stroke actions across various trajectories performed by players. Daniel Peralta³⁵ utilized CNNs based on acceleration data from IMUs to classify seven types of forehand strokes and two states in five amateur athletes. The classification accuracy for the five athletes was 96.73%, 89.08%, 93.02%, 90.53%, and 95.63%, with an average accuracy of 93.00%. When using both acceleration and gyroscope data, the average accuracy increased to 95.91%. Tim Steels and his team³ placed IMUs on the wrist, upper arm, and racket handle, using acceleration data from these locations to classify seven types of stroke actions using CNN. When using only accelerometer data at a sampling frequency of 50 Hz, the accuracy was 86%. By combining accelerometer and gyroscope data, the accuracy improved to 99%, with weighted accuracy rates of 93% and 96% for the wrist and upper arm, respectively. This is just slightly the 98% accuracy achieved with sensors mounted on the racket. Indrajeet Ghosh et al.¹⁹ used data from IMUs on four body parts and applied four machine learning methods to identify strokes and lower limb movements of athletes at different levels. Based on the recognition of 12 types of strokes, they designed an error learning index module, which calculated the error between the ideal posture of professional athletes and that of others and used this error as a scoring index for competition. In our study, we controlled two IMUs using the ESP32, which, while sacrificing some portability, significantly improved the accuracy of the data in the time domain and reduced the adverse effects of subsequent data interpolation. Ben Van Herbruggen et al.¹⁸ have made notable progress in the field of badminton skill and tactical recognition. In their study, they utilized two IMUs and an ultra-wideband (UWB) positioning system, allowing real-time tracking of players' positions during matches and recognition of various shot types. By using CNN to classify 13 different stroke actions, the team achieved an accuracy rate of 85% with only IMUs data. Combining it with UWB positional data, the accuracy of shot classification increased to 88.99%. Additionally, they conducted a classification study on the tactics used by players during matches. When relying solely on UWB data for four tactical categories, the accuracy was relatively low at only 53.5%. However, by integrating shot action data with UWB position information, the accuracy for tactical classification significantly increased to 80%. In the study by Z. Wang³⁶, four IMUs data sets were used in combination with Hidden Markov Models (HMM) to classify 14 badminton hitting actions. Chang Tai Kiang highlighted¹⁰ in his article that using only accelerometers is insufficient to evaluate badminton smash performance. He developed a localized sensing system that integrates acoustic emission sensors with accelerometers mounted on the racket head, allowing for a more accurate analysis of additional information related to overhead smash techniques and offering insights into performance metrics for badminton smashes.

Srimath et al. reported that the 1D-CNN model achieves higher classification accuracy than the Bi-Directional Long Short-Term Memory (Bi-LSTM) model in RGB-based human activity recognition tasks. This improvement was achieved without relying on depth or multi-modal data³⁷. Zhang et al. selected the 1D CNN

Study	Goals	Methods	Accuracy
³⁴	Shot recognition	RF, SMO, SMO, RBF, NB	84%
³⁵	Stroke Action and State recognition	CNN	93%
³	Shot and activity recognition	CNN, DNN	98%
¹⁹	Shot classification, performance predictor	CNN, KNN	89%
¹⁸	Shot and strategy recognition	CNN, LSTM	90.0% and 80%
³⁶	Shot classification	HMM	96%
This study	Shot classification, Shot trajectories	1D-CNN	97.16% and 86.07%

Table 1. Existing research on badminton stroke recognition.

+ LSTM algorithm as the optimal method based on experimental analysis and comparison, and constructed a human motion behavior recognition network model to process data collected by smart wearable devices³⁸. In the field of basketball, researchers developed a shooting posture recognition system based on a 1D-CNN, which collects motion data from the dominant hand and foot through sensor fusion. This system achieved an accuracy of 98.6%, which further demonstrates the applicability and robustness of CNN architectures in sport-specific motion classification tasks³⁹. Similarly, Ma et al.⁴⁰ mounted an IMU on the athlete's right wrist and employed a GoPro camera to capture synchronized video data, enabling the recognition of nine fundamental basketball movements. Their neural network-based model achieved an accuracy of 98.9% in experiments with a single subject, validating the feasibility of the method under controlled conditions. These findings demonstrate that integrating multiple sensor data sources can significantly enhance the accuracy of badminton skill and tactical recognition, providing strong support for sports training and match analysis. The reviewed literature indicates that machine learning and deep learning have been successfully applied to motion recognition in badminton, with promising results. As a sport, badminton is characterized by rapid changes, a fast pace, and high speed. To achieve excellent performance, players need advanced technical and tactical abilities. In competitive play, the ability to adjust shot trajectories according to varying situations is a key component of a player's proficiency. Previous studies have mainly focused on recognizing various stroke actions but have lacked an analysis of stroke actions across different trajectories. In this context, our study primarily focuses on utilizing neural networks and IMUs to recognize athletes' stroke actions across diverse shot trajectories. Specifically, we leverage 1D-CNN and IMUs data to identify various stroke actions while also recognizing various shot trajectories, providing a more scientific reference for participants of varying skill levels.

Application of ESP32 in motion recognition

Previous studies on badminton action recognition have employed multiple sensors to capture athletes' strokes. However, using multiple sensors for data collection presents a challenge in synchronizing their timing accurately. This often requires post-processing methods, such as interpolation, to align the data. Although interpolation can help synchronize the data, it may distort the stroke information, increase computational costs, and add complexity to the training model. With advancements on the Internet of Things (IoT) and the growing demand for miniaturized devices, Espressif Systems introduced the ESP32 microcontroller in a QFN48 package in September 2016, designed to replace the earlier ESP8266 microcontroller⁴¹. The ESP32 is a powerful microcontroller featuring built-in Wi-Fi and Bluetooth, making it an ideal solution for IoT applications. It includes two independently operating Xtensa LX6 CPU cores, each with a maximum frequency of 240 MHz, and supports external crystal oscillators, four 64-bit general-purpose timers, a PWM timer, an RTC clock, and various low-power clock modes⁴². Its dual-core architecture enables parallel task processing, and the timers can count up or down, freeze, or trigger events. Considering the powerful timing capabilities of the ESP32, previous studies have proposed a wireless timing system based on the ESP-to-ESP communication protocol to overcome the limitations of Bluetooth's short-range and the dependence on internet connectivity. This system enables remote activation of the timer and real-time data transmission, while allowing the timer to be automatically stopped by a terminal sensor at the end of a competition, thereby significantly reducing human-induced timing errors and improving measurement accuracy⁴³. To enhance training efficiency and competitive performance in ice and snow sports, Saavedra et al.⁴⁴ developed a data acquisition system based on ESP32 modules and IMUs to enable real-time monitoring of athletes' limb acceleration data. In addition, a supporting Android application was developed to facilitate data interaction and visualization, further validating the integration capability and application potential of ESP32 technology in the field of sports science. Building upon this, K. Ishida et al.²⁴ proposed a sports framework that integrates IoT applications with sports skills research, emphasizing wearable devices with inertial sensors. In summary, advancements in IoT technology and microcontrollers such as the ESP32 have opened up new possibilities for motion recognition and data acquisition in sports. ESP32-based platforms enable more accurate and real-time data collection through wireless communication and integrated sensing. These systems also reduce computational load, minimize human intervention, and expand the scope of biomechanical analysis. Their feasibility and application potential have been demonstrated in fields such as badminton and winter sports, laying a solid foundation for the development of efficient, low-power, and scalable motion monitoring solutions.

Model construction and data processing

Data information and 1D-CNN

As shown in Fig. 1, to facilitate learning of the feature for different stroke actions, the 1D-CNN model was built to recognize the six badminton stroke actions. Two IMUs were employed in this process, positioned at the distal end of the wrist radius and the lower end of the badminton racket handle to capture motion data from both segments¹⁹. The IMUs at the wrist radius measure angular changes in the wrist joint during the strike, while the other IMUs at the racket handle record the angular variations of the racket during impact. To acquire more precise data, the ESP32 microcontroller was used to perform time synchronization of the signals from the two IMUs, forming a multi-sensor data acquisition and synchronization system. The microcontroller, using the external crystal oscillator and timers embedded in it, was employed to synchronize events in real time. This design ensures that during an athlete's stroke motion, the triaxial acceleration values remain synchronized within the same time frame, effectively avoiding data distortion and minimizing the computational load that would otherwise arise from post-processing synchronization. The captured motion data were merged in chronological order to create a fused dataset. Specifically, the data from the wrist IMUs and the racket IMUs are represented as $S(M)$ and $S(\Theta)$, respectively. The multi-sensor data acquisition and synchronization method performs time synchronization and data merging, allowing for comparison and analysis of different badminton

strokes within the same time dimension. This approach also enhances the ability of the 1D convolutional neural network classifier to recognize various stroke types and trajectories.

To eliminate the dimensional differences between the two IMUs, we apply normalization using a formula. F_j is the j -th element of F (where $j = 1, 2, 3, \dots, N$), and N represents the dimensionality of the F dataset. The functions $\min$ and $\max$ represent the minimum and maximum values of the variables across all the badminton Stroke actions, respectively. Stroke events were segmented using a sliding-window approach, where each window consisted of 100 samples with a stride of 50 samples, resulting in 50% overlap between consecutive windows. The true stroke actions are manually labeled, with 80% of the dataset used for training and 20% for prediction.

$$F_j = (F_j - \min F) / (\max F - \min F) \tag{1}$$

The feature maps from the two IMUs are concatenated to form an input of size $18 \times 1 \times 1$. The 1D-CNN used in this study is shown in Table 2, consisting of two convolutional blocks with ReLU activations and batch normalization, one max-pooling layer, a fully connected layer, and a Softmax classifier. Specifically, Conv1 uses 8 filters with a 3×1 kernel, followed by batch normalization and ReLU; a 2×1 max-pooling layer is then applied to reduce the feature dimension. Conv2 uses 16 filters with a 3×1 kernel, followed by batch normalization and ReLU. The pooled features are flattened and fed to a fully connected layer with six neurons, followed by Softmax to produce class probabilities. The network was trained with cross-entropy loss and the Adam optimizer (learning rate = 0.005, batch size = 64, iterating over the 500 epochs). In preliminary pilot experiments on a separate development set, this configuration yielded stable convergence and strong validation accuracy; therefore, we fixed it as the default in the main experiments and did not perform hyperparameter search on the study dataset. Early stopping was used with a patience of 10 epochs. Stratified 5-fold cross-validation was used for performance estimation; preprocessing was fitted on the training portion only and applied to the validation portion. Dropout was not used due to the lightweight architecture and the regularization provided by batch normalization and early stopping.

To quantify the performance of stroke action recognition, the metric of accuracy was defined.

$$\text{Accuracy} = (TP + TN) / (TP + FP + TN + FN) \tag{2}$$

Among them, TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

To create a multi-sensor data acquisition and synchronization system using the ESP32 and IMUs, we assign T^1 and T^2 represent $S(M)$ and $S(\Theta)$ from the two IMUs respectively. To ensure synchronization of the data in the time sequence, using system as a reference and calculate the time delay Δt between the two IMUs.

$$\Delta t = T^2 - T^1 \tag{3}$$

By using the value of Δt the delayed signal in the timestamp is compensated. T_2^0 represents the timestamp of $S(\Theta)$ after compensation.

$$T_2^0 = T^1 + \Delta t \tag{4}$$

Let S_{fused} represent a set, where the two time-synchronized signals $S(M)$ and $S(\Theta)$ are fused.

$$S_{\text{fused}} = [S(M), S(\Theta)] \tag{5}$$

Different stroke actions and stroke trajectories

By analyzing the connection between unforced mistakes, successful shots, and match results, we can gain a deeper understanding of how crucial actions, such as unforced mistakes and scoring points, influence the final outcome⁴⁵. Research has also shown that a significant number of unforced mistakes often contribute to losing a match or set. Consequently, we divided the research objectives into two parts: (1) identifying six basic badminton techniques, and (2) identifying six different shot trajectories based on these techniques. Using two IMUs controlled by an ESP32 microcontroller, we first collected data on the six basic badminton techniques. These actions include:

Layer No.	Layer Description	Output Size*
1	Input: $18 \times 1 \times 1$	$18 \times 1 \times 1$
2	Conv1: 8 filters, kernel 3×1 , stride 1, ReLU	$18 \times 1 \times 8$
3	MaxPool1: pool 2×1 , stride 2×1	$9 \times 1 \times 8$
4	Conv2: 16 filters, kernel 3×1 , stride 1, ReLU	$9 \times 1 \times 16$
5	Fully Connected: 6 units	-
6	Softmax + Classification	-

Table 2. 1D-CNN network architecture. * Output shown as $H \times W \times C$; here $W=1$ because convolution is 1-D along the feature axis.

- 1) Forehand clear(FC): This overhead stroke technique is used to send the shuttle high and deep into the opponent's backcourt, giving the player enough time to recover and transition into a defensive position.
- 2) Forehand drop(FD): This overhead stroke technique is used to send the shuttle from the backcourt to the opponent's front court with a low, close trajectory, landing near the net. This technique is somewhat offensive in nature.
- 3) Forehand smash(FS): The forehand smash, a highly offensive overhead stroke technique, drives the shuttle from the backcourt to the opponent's mid- or front-court with a fast, low trajectory, often catching the opponent off guard.
- 4) Forehand lift(FL): This defensive underhand stroke sends the shuttle from the front court to the opponent's backcourt with a high, deep trajectory. It gives the player time to recover and reposition for defense, while also forcing the opponent to move.
- 5) Forehand net tumble(FNT): A control-oriented defensive technique performed from a low position, in which the player gently sends the shuttle over the net, aiming to land it close to the opponent's front court. This forces the opponent to move closer to the net, limiting their shot options.
- 6) Forehand cross-net shot(FNS): A common net-front offensive and defensive technique, the shuttle is directed across the net from one side to the other, often to catch the opponent off balance.

Players must quickly and accurately assess the movement direction, speed of the shuttlecock, and the opponent's intentions⁴⁶. This ability is crucial for determining the outcome of high-level matches. Different shot trajectories form the tactical strategies employed by athletes, making them key components in achieving victory. Shot trajectories are primarily categorized into straight and diagonal lines. However, when combined with various shot techniques, a wide variety of distinct shot trajectories can emerge. Given that the straight shot from an overhead position is essentially the same as the forehand straight shot technique, and the net shot technique has minimal impact on the shuttlecock's trajectory, these are excluded from consideration. The above fifteen different striking trajectories are displayed in the following Fig.2 Based on different shot techniques; the following fifteen distinct shot trajectories are identified:

- 1) Forehand Straight Clear (FSC): Using the clear shot technique, the shuttle is hit with a high and deep straight trajectory, targeting the opponent's backcourt. The shuttle travels from the player's forehand area to the opponent's overhead area.
- 2) Forehand Straight Drop Shot (FSD): Using the drop shot technique, the shuttle is hit with a flat and fast straight trajectory from the rear court, landing in the opponent's front court. The shuttle travels from the player's forehand rear-court area to the opponent's backhand front-court area.
- 3) Forehand Straight Smash (FSS): Using the smash technique, the shuttle is hit with a fast and downward straight trajectory, aimed at the opponent's mid-front court. The shuttle travels from the player's forehand rear-court area to the opponent's backhand mid-front court area.
- 4) Forehand Cross-Court Clear (FCC): Using the clear shot technique, the shuttle is hit from the player's rear forehand area with a high and deep cross-court trajectory, traveling from the player's rear court to the opponent's diagonal rear court.
- 5) Forehand Cross-Court Drop Shot (FCD): Using the drop shot technique, the shuttle is hit with a flat and fast cross-court trajectory from the player's rear forehand area, landing in the opponent's forehand front court area.
- 6) Forehand Cross-Court Smash (FCS): Using the smash technique, the shuttle is hit with a fast and steep cross-court trajectory from the player's rear forehand area, targeting the opponent's forehand mid-front court area.
- 7) Overhead Cross-Court Clear (OCC): Using the clear shot technique, the shuttle is hit from the player's rear overhead area with a high and deep cross-court trajectory, aiming for the opponent's rear overhead area.
- 8) Overhead Cross-Court Drop Shot (OCD): Using the drop shot technique, the shuttle is hit from the player's rear overhead area with a flat and fast cross-court trajectory, landing in the opponent's front backhand area.
- 9) Overhead Cross-Court Smash (OCS): Using the smash technique, the shuttle is hit with a fast and steep cross-court trajectory from the player's rear overhead area, targeting the opponent's backhand mid-front court.
- 10) Forehand Cross-Court Lift (FCL): Using the lift shot technique, the shuttle is lifted from the player's forehand frontcourt area with a high, arcing cross-court trajectory, reaching the opponent's forehand rear court.
- 11) Backhand Cross-Court Lift (BCL): Using the lift shot technique, the shuttle is lifted from the player's backhand frontcourt area with a high, arcing cross-court trajectory, reaching the opponent's overhead rear court.
- 12) Forehand Net Cross Shot (FNC): Using the net cross shot technique, the shuttle is hit from the player's forehand frontcourt area, crossing to the opponent's forehand frontcourt.
- 13) Backhand Net Cross Shot (BNC): Using the net cross shot technique, the shuttle is hit from the player's backhand frontcourt area, crossing to the opponent's backhand frontcourt.
- 14) Backhand Drop Shot (BD): Using the drop shot technique, the shuttle is hit from the player's rear-court area with a back-facing posture, traveling over the head to the opponent's front court.
- 15) Backhand Clear Shot (BC): Using the clear shot technique, the shuttle is hit from the player's rear-court area with a back-facing posture, traveling over the head to the opponent's rear court.

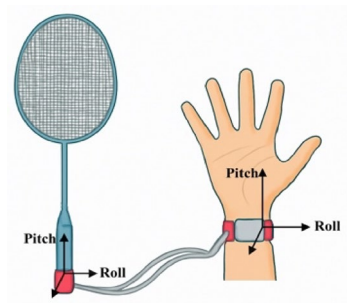


Fig. 3. Wearable Device Usage Method in Badminton.

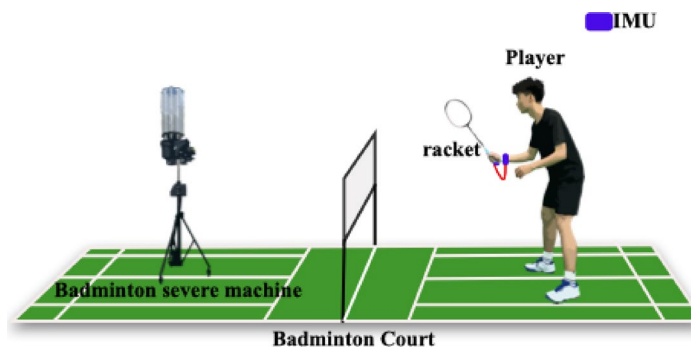


Fig. 4. Badminton stroke experience setup.

Experiment

Hardware platform

Figure 3 illustrates the placement of two sensors during the striking action. In upper-limb-dominant tasks, energy generation and transfer generally follow a proximal-to-distal sequence. Therefore, the positioning of the two IMUs must consider this kinetic chain relationship, forming an “upper limb–wrist–racket” motion chain⁴⁷. IMU-based motion analysis typically relies on wearable devices with embedded sensors to record motion and orientation data⁴⁸. When mounted on sports implements, the sensor devices should be non-invasive to the human body. Referring to the sensor placement strategies in table tennis⁴⁹, as well as previous IMU applications in badminton¹⁹, tennis⁵⁰, and basketball⁵¹, we chose to place the IMUs at the end of the badminton racket and on the radial side of the athlete’s racket-holding wrist. The nine-axis IMUs (HWT906) are compact in size ($25 \times 25 \times 1.6 \text{ mm}^3$) and have low power consumption (about 0.1 watts), with measurement ranges of $\pm 160 \text{ m/s}^2$ for acceleration, $\pm 2000 \text{ deg/s}$ for angular velocity, and $\pm 4900 \mu\text{T}$ for the magnetometer. The axis calibration follows the right-hand rule. Specifically, the X-axis of the wrist-radius IMUs corresponds to the direction of the thumb, and the Y-axis corresponds to the direction of the right-hand index finger. The IMUs placed at the bottom of the badminton racket handle have their Y-axis corresponding to the racket’s face and their X-axis corresponding to the side of the handle. The entire sensing system is powered by a 5-volt battery and simultaneously collects signals from the two IMUs via the SPI bus protocol at a sampling frequency of 100 Hz. To ensure the stability of the serve, we used the SIBOASI smart serving machine (SS-S4025A), which has a maximum input power of 360 watts, a serving frequency ranging from 1.2 to 5.5 s, and an elevation angle between -15 and 33 degrees. It was connected to a mobile app to adjust the serving strategy based on different hitting positions, ensuring the quality of the serve.

Experimental protocol

In this experiment, six male national-level first-class athletes, all of whom had not been injured in the past six months, were recruited (average age: 22 ± 3 years, height: 176 ± 8 cm, and weight: 70 ± 6 kg). All subjects were right-handed and participated in the experiment with informed consent. All striking action tests were conducted in an indoor badminton court. As shown in Fig. 4, the athlete wears the sensors and stands in the center of the badminton court, with the serving machine positioned on the opposite side. Holding the badminton racket, the athlete waits to strike. The serve is directed towards the player at a frequency of 4 s, allowing ample time for the player to execute their action and prepare for the subsequent shot. Meanwhile, the two sensors are fixed at the end of the racket handle and on the inner side of the right radius using athletic tape and nylon straps, respectively. The participants were required to complete six different striking actions and fifteen different shot trajectories, with each action performed 20 times with valid results. Before each experiment, the IMUs were restarted and left stationary for calibration. A 2-minute rest period was set between each pair of striking actions to ensure recovery of the athlete’s energy system and to prevent muscle fatigue from negatively impacting the striking actions. All

methods were carried out in accordance with relevant guidelines and regulations. All experimental protocols were approved by the Ethics Committee of Chengdu University.

Result

In this section, all models were run in MATLAB (MathWorks R2022b) on the same computer with an Apple M2 chip, integrated GPU, and 16 GB unified memory. The classification results are presented using heatmaps, t-SNE plots, and confusion matrices. Since angular velocity is associated with increased coactivation of antagonistic muscles⁵², it is further utilized to perform a more detailed analysis of the misclassified results.

Initial analysis multi-sensor data acquisition and synchronization feature

Figure 5a and b collectively present 12 heatmaps corresponding to six types of striking actions, each representing data from two IMUs. Each IMU collects nine feature values, reflecting acceleration, angular velocity, and angle data along the X, Y, and Z axes. These feature values vary across different striking actions, each displaying distinct signals. $S(\Theta)$ reflects the data from the IMU located below the racket handle, while $S(M)$ reflects the data from the IMUs at the wrist. The horizontal axis represents the different stages of the striking action, from 0% to 100%, and the vertical axis represents the nine elements captured by the IMUs. For example, in the eighth and ninth rows for FC and FS, the racket and wrist movement amplitudes differ significantly, whereas in FD, the variations are smaller. FNT shows even smaller movement amplitudes and smoother variations in the first and second rows. Furthermore, within the same striking action, the heatmaps from the two IMUs complement each other. Before inputting the data into the striking action recognition model, Kalman filtering is applied to the raw data. This step effectively reduces noise, improves feature extraction, and enhances the model's generalization ability.

To better visualize the feature extraction of different badminton stroke actions by the 1D CNN, we use t-SNE plots to project both the raw data and the extracted features onto a two-dimensional plane for comparison. As shown in Fig. 6, the perplexity is set to 20, with the default Euclidean distance metric used to measure the distances between values, and the learning rate is set to 200. The extracted features are more distinguishable than the original data, particularly when incorporating the complementary information of $F^{S(\Theta)}$ and $S(M)$, which enhances the clarity of the class boundaries and reduces the confusion present in the original features. Compared to $F^{S(M)}$, $F^{S(\Theta)}$ as a single signal contributes more significantly to the clarity of the class boundaries. Notably, the confusion between the FC and FS classes is significantly reduced, and the confusion between FD, FC, and FS classes is also minimized. The FNT class, due to its smaller overall motion range, appears more scattered in

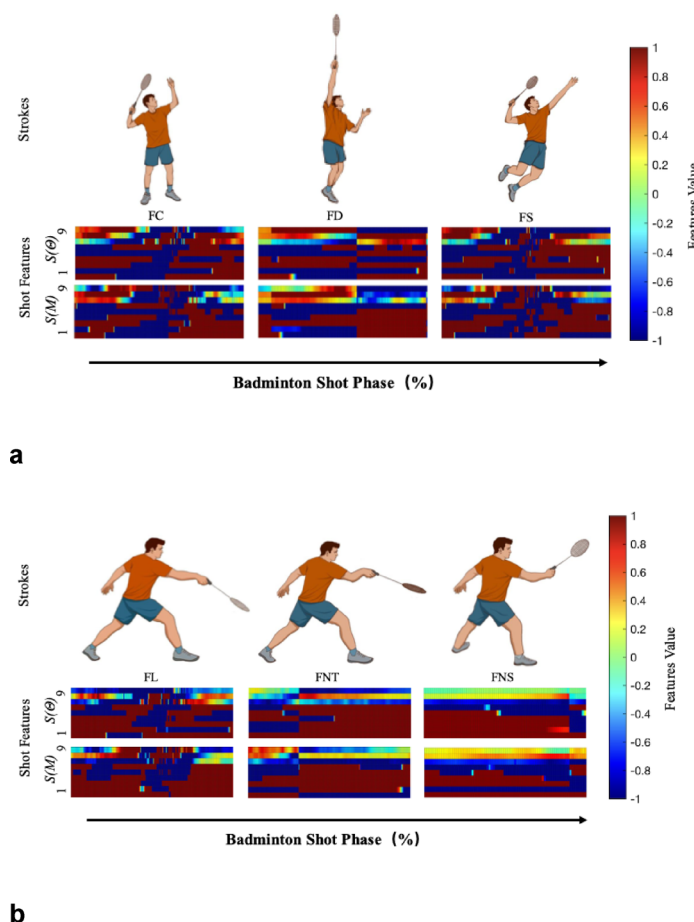


Fig. 5. Heat maps of three different stroke movements within one movement cycle.

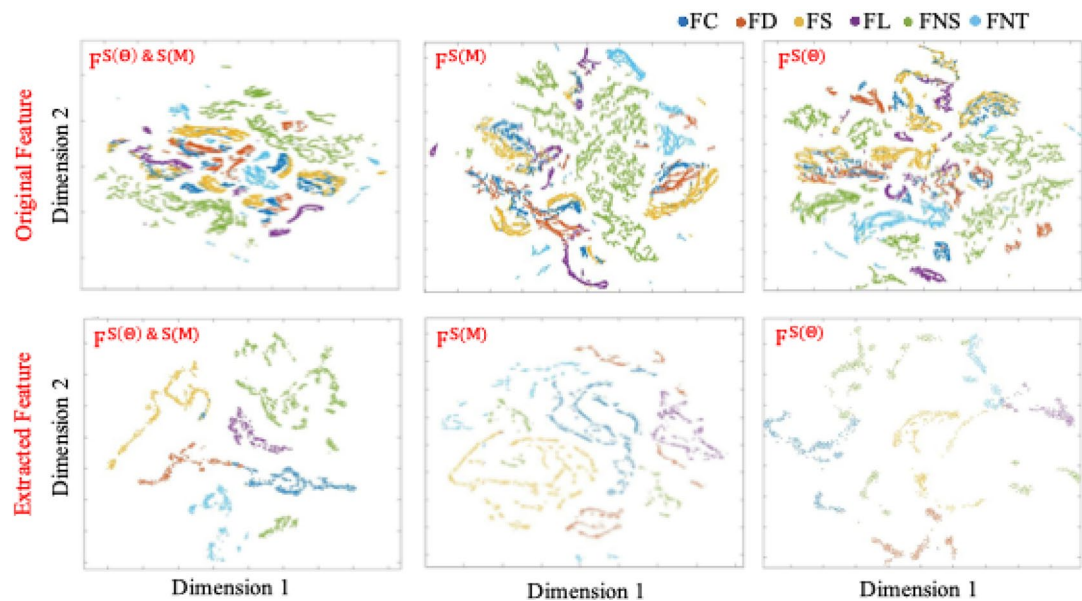


Fig. 6. t-SNE visualization for six badminton stroke actions.

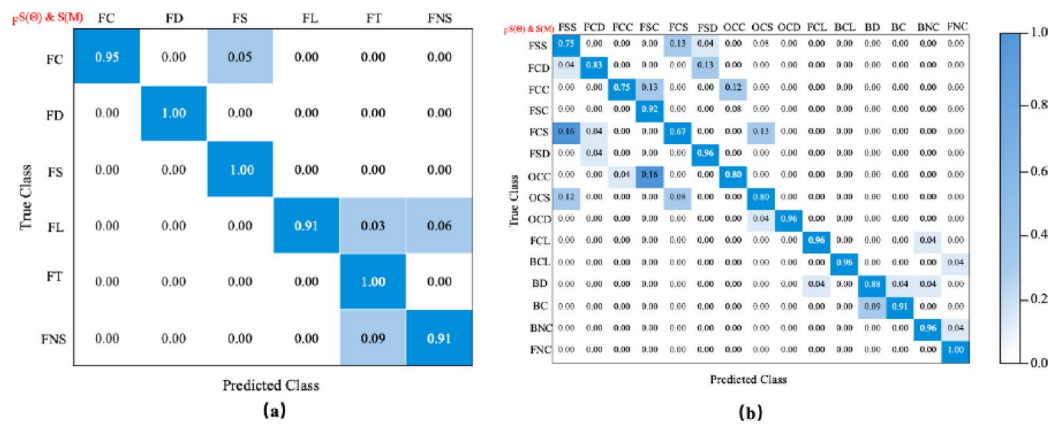


Fig. 7. Confusion Matrixes of 1D-CNN for (a) Six human stroke actions; (b) Fifteen badminton trajectories.

the plot. Subsequently, the extracted features are accurately classified into six stroke actions through six fully connected layers and a SoftMax activation function.

Using CNN to recognize different badminton stroke actions

Based on the t-SNE plot, we analyzed the classification performance of the 1D-CNN model in recognizing six types of badminton strokes. With multi-sensor data acquisition and synchronization, the model achieved an overall classification accuracy of 97.16%. As shown in Fig. 7a, the confusion matrix indicates that the model has some difficulty distinguishing between FL and FNS. This is mainly because, during the pre-hit phase, athletes often adopt similar racket face angles to maintain consistency in their shots, making the two actions appear highly similar. In addition, individual differences in how players execute these strokes may further contribute to the confusion. Overall, this phenomenon is consistent with our expectations of the model's recognition performance.

In actual training, coaches can address the issues in players' shots based on the classification results. Specifically, when there is some confusion between the FC and FS, the coach, based on the classification results, can improve the stroke point or strengthen the coordination of muscles from the legs to the wrists according to the spatial height and the characteristics of technical movements. This is done in consideration of individual player differences. Targeted training for the two stroke actions is then implemented in subsequent training sessions, using the stroke data collected by IMUs. In terms of the training plan, based on the results of the confusion, the training intensity and measurement for different technical movements should be reasonably allocated. The intensity and measurement should be appropriately reduced for excellent shot types, while adjustments should be made for poor shot types.

Trajectories	Median Peak Value (X- Axis)	Median Peak Value (Y- Axis)	Median Peak Value (Z- Axis)
FSS/FCS	1602.9/1514.4	1649.6/1657.0	1975.3/1999.8*
FSS/OCS	1602.9/1368.1	1649.6/1394.2*	1975.3/1885.5
FCS/OCS	1514.4/1368.1*	1657.0/1394.2*	1999.8/1885.5*

Table 3. The Mann-Whitney U test was used to compare the peak angular velocities of FSS, FCS, and OCS along the X, Y, and Z axes. (deg/s). *represents $p < 0.05$, indicating statistical significance.

Trajectories	Median Peak Value (X- Axis)	Median Peak Value (Y- Axis)	Median Peak Value (Z- Axis)
FSS/FCS	640.2/681.4	653.0/703.8	625.4/664.6
FSS/OCS	640.2/564.9*	653.0/581.7*	625.4/555.3
FCS/OCS	681.4/564.9*	703.8/581.7*	664.6/555.3*

Table 4. The Mann-Whitney U test was used to compare the RMS angular velocities of FSS, FCS, and OCS along the X, Y, and Z axes. (deg/s). *represents $p < 0.05$, indicating statistical significance.

Using CNN to recognize different stroke trajectory actions

In the previous section, we achieved high-accuracy recognition of six stroke actions. In this section, we further classified stroke trajectories within the same technical actions, achieving an 86.07% recognition rate. As shown in Fig. 7b, the recognition accuracy of FCS was relatively low, with substantial confusion observed among FSS, FCS, and OCS. For FCC and OCC, some samples were misclassified, with FCC occasionally recognized as FSC or OCC, and OCC sometimes misclassified as FSC. Nevertheless, the CNN demonstrated superior performance in recognizing net play techniques, with FCL, BCL, and net kills all achieving high accuracy. Overall, the CNN showed strong classification capability across most technical actions.

To further examine the cause of confusion between the FSS, FCS, and OCS stroke trajectories, we selected the $F^{S(M)}$ data from the wrist and performed a Mann-Whitney U test on the Peak angular velocities of these actions across different axes. As shown in Table 3, when comparing FSS and FCS, only the Z-axis peak differed significantly. Both strokes exhibit nearly identical kinematics in the X–Y plane during preparation, impact, and follow-through, resulting in similar X- and Y-axis peaks. However, FCS demands a larger whole-body rotation—driven by powerful leg extension, torso twist, and coordinated shoulder–elbow–wrist motion—which amplifies rotation about the Z-axis. In contrast, FSS and OCS diverges solely along the Y axis: executing the diagonal overhead chop requires greater racket-face abduction/adduction, producing a higher Y-axis peak without affecting X- or Z-axis rotations. Finally, FCS and OCS differ markedly along all three axes. Their diagonal impact trajectories increase Y-axis rotation, while FCS’s emphasis on explosive sagittal-plane drive heightens X-axis peaks. Moreover, variations in the timing and magnitude of trunk and arm rotations generate distinct Z-axis profiles. These findings not only corroborate prior kinematic studies of elite players but also underscore the importance of tailored training interventions that address the unique rotational demands of each stroke.

Table 4 presents the results of Mann–Whitney U tests comparing angular-velocity RMS values among the three stroke types. When comparing FSS and FCS, no significant differences emerged on the mediolateral (X), anteroposterior (Y), or vertical (Z) axes, indicating that these strokes share nearly identical global kinematic profiles. This kinematic homogeneity likely underlies the occasional confusion by the CNN, which struggles to distinguish between such similar movement patterns. In contrast, FSS and OCS differed significantly on the X and Y axes: FSS showed stronger sagittal-plane oscillations throughout the stroke cycle, whereas OCS combined comparable sagittal intensity with additional frontal-plane rotation; RMS values on the Z axis remained similar. All three axes exhibited significant RMS distinctions between FCS and OCS. FCS stroke trajectory diagonal trajectory generates continuous planar force coupled with rhythmic “open–close” racket-face motions about the Z axis, while OCS follows a dominant overhead-downward path with sustained torsional intensity. Despite these clear biomechanical differences, residual CNN confusion may stem from inter-athlete variability.

In training, the classification results of different stroke trajectories are not only used to improve the athlete’s technical errors and adjust their training plan but can also be combined with the athlete’s personality traits to develop their tactical style. By observing the classification results of an athlete’s different stroke trajectories, well-performing trajectories can be combined in various ways, and the use of poorly-performing trajectories in the tactical strategy can be reduced. Combining the well-performing stroke trajectories with the athlete’s personality traits helps formulate tactical decisions that align with the athlete’s individual characteristics. Finally, this helps athletes improve training efficiency, enhance competition performance, and achieve excellent sports results.

Under stratified 5-fold cross-validation, the CNN model achieved high and stable performance. For stroke classification, the per-fold accuracies were 96.30%, 96.90%, 97.20%, 97.60%, and 97.80%, yielding an overall mean accuracy of $97.16\% \pm 0.6\%$. For trajectory classification, the accuracies were 84.55%, 85.55%, 86.10%, 86.80%, and 87.36% across the five folds, resulting in a mean of $86.07\% \pm 1.1\%$. As shown in Fig. 8(a), the CNN significantly outperformed other methods such as LDA, LRA, SVM, DTA, and KNN, demonstrating superior performance and stability in motion recognition tasks. However, in terms of training time, the CNN required considerably more time, taking 133 s. By contrast, as shown in Fig. 8(b), KNN achieved the shortest training time, requiring only 27 s.

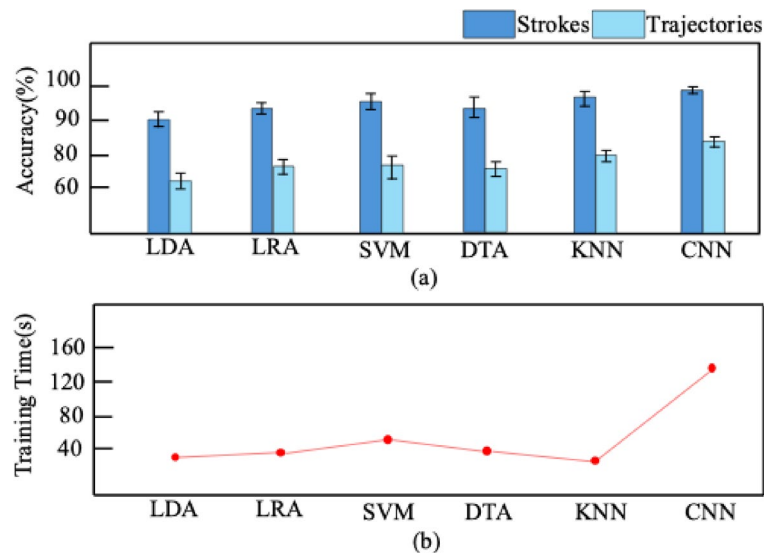


Fig. 8. (a) 5-Fold Cross-Validation Results and (b) Model Training Time.

Classifier	Feature	Accuracy actions	Accuracy trajectories
LDA	$F^{S(\Theta)} \text{ and } S(M)$	90.15	62.85
	$F^{S(M)}$	80.04	61.79
	$F^{S(\Theta)}$	88.13	69.38
LRA	$F^{S(\Theta)} \text{ and } S(M)$	92.47	75.54
	$F^{S(M)}$	91.67	68.72
	$F^{S(\Theta)}$	91.03	71.35
SVM	$F^{S(\Theta)} \text{ and } S(M)$	94.32	75.06
	$F^{S(M)}$	85.81	66.92
	$F^{S(\Theta)}$	92.75	66.01
DTA	$F^{S(\Theta)} \text{ and } S(M)$	92.81	72.93
	$F^{S(M)}$	90.12	50.77
	$F^{S(\Theta)}$	89.12	57.24
KNN	$F^{S(\Theta)} \text{ and } S(M)$	94.94	78.71
	$F^{S(M)}$	91.67	59.39
	$F^{S(\Theta)}$	92.11	68.21
CNN	$F^{S(\Theta)} \text{ and } S(M)$	97.16	86.07
	$F^{S(M)}$	93.32	71.28
	$F^{S(\Theta)}$	94.24	73.58

Table 5. The accuracy of different classifiers for badminton stroke action and trajectories recognition (%).

Robustness of Multi-Sensor data fusion to various algorithms

We used five other classifiers, including Decision Tree Analysis (DTA), support vector machines (SVM), decision tree analysis, K-nearest neighbor (KNN), logistic regression analysis (LRA), and linear discriminant analysis (LDA) to verify the generalization ability of data merging. Table 4 shows the accuracy of six classifiers in the identification of badminton stroke actions and stroke trajectories, among which the 1D-CNN demonstrates superior performance, reaching 97.16% and 86.07% in $F^{S(\Theta)} \text{ and } S(M)$ the features, respectively. Under the three characteristics of stroke actions and trajectories, the 1D-CNN achieved the best results, showing excellent processing ability for different data types.

Discussion

This study combines multi-site sensor information fusion with 1D-CNN to recognize different badminton stroke actions and flight trajectories. This is the first time that a multi-sensor data acquisition and synchronization system, consisting of ESP32 and IMUs, has been used to monitor badminton movements. The following is a further discussion of the above aspects.

- 1) As shown in Table 5, previous related studies have either used a single sensor or an excessive number of sensors. A single sensor may not fully capture the complete data of badminton stroke actions, while using too many sensors may reduce the comfort of wearable devices. By using two sensors—one on the wrist and one on the racket—complementary data on human motion can be captured more comprehensively than with a single sensor, achieving accuracies of 97.16% for stroke actions and 86.07% for stroke trajectories.

Study	Sensor Configuration	Accuracy
34	One IMU	84%
35	One IMU	93%
19	Four IMUs	89%
18	One IMU and UBW	90.0%
This study	Two IMUs and ESP32	97.16%

Table 6.. Comparisons with Related Badminton stroke Recognition Studies.

- 2) In practical applications, using a two-sensor configuration not only ensures high recognition accuracy but also balances comfort by reducing the discomfort caused by sensor rigidity, allowing athletes to perform movements more comfortably. This, in turn, enhances data collection quality and improves the model's performance in classifying different stroke actions.
- 3) In Table 4, the comparison of various machine learning methods shows that CNN outperforms other methods in recognizing stroke actions and trajectories, achieving accuracies of 97.16% and 86.07%, respectively, and also demonstrates the best performance across comparisons of $F^{S(M)}$, $F^{S(\Theta)}$, and $F^{S(\Theta)}$ and $S(M)$ signals. However, considering that CNN requires feature extraction and learning through multiple convolutional layers, it results in a relatively higher computational load and longer training time. Therefore, during model training, CNN takes longer to train compared to models such as KNN, SVM, and DTA.
- 4) Compared to the UWB ultra-wideband system, the ESP32 offers better cost-effectiveness and, when combined with the simpler 1D-CNN, holds greater potential for commercial deployment. Considering the portability required for wearable devices, the ESP32 also has an advantage in terms of timely hardware deployment.
- 5) The feature variations across different stroke actions are visually represented and more intuitively referenced in the heatmaps presented in Fig. 5a and b, the t-SNE plot in Fig. 6, and the peak and RMS values in Tables 3 and 6.. These provide important theoretical support for enhancing badminton training.
- 6) Although the dual-IMU configuration with the ESP32 has improved recognition accuracy to some extent, confusion remains between certain actions, especially when the movement patterns are similar. Future research could further enhance the model's accuracy and generalization by exploring different sensor types, adopting more advanced data fusion techniques, and investigating more detailed feature extraction methods. Potential limitations should also be acknowledged, including data drift in IMUs during long-term use and variations in model performance under different application conditions. Future work will address these issues more comprehensively to strengthen the model's reliability and applicability in real-world scenarios.
- 7) This study successfully achieved the research objective of stroke action classification. However, the absence of female athletes limits the evaluation of skill level and gender effects on model performance. Moreover, age and body shape differences among athletes should also be considered. To address these limitations, future research will expand the sample size to include athletes of different genders, age groups, skill levels, and body types. This will allow a more comprehensive validation of the model's generalizability and accuracy, particularly where gender differences may influence movement pattern recognition, thereby enhancing its applicability across a broader population.

Conclusion

Based on wearable sensing and machine learning techniques, this study achieves accurate recognition of six badminton stroke techniques and fifteen stroke trajectories. Experiments were conducted with six national-level badminton athletes, and the wearable system consisting of two IMUs, simultaneously captures the motions of the badminton racket and human wrist. To establish the mapping between raw signals and motion patterns, the IMUs measurements are fed into a constructed 1D-CNN for training and prediction. To intuitively display feature differences in different patterns, heatmaps and t-SNE plots are used, which support the results achieved by 'black box' machine learning models. The effectiveness of the proposed method is demonstrated by the pattern recognition accuracy of 97.16% for six badminton stroke motions and 86.07% for fifteen badminton flight trajectories. This study provides an effective method for badminton motion monitoring, which is expected to be valuable for enhancing badminton training to improve athletes' competitive skills.

These results suggest that with proper data collection and neural network training, the model can effectively manage the complexity involved in classifying both actions and trajectories. However, the CNN's limitations in capturing temporal features led to minor confusion when classifying certain shot trajectories. In future research, we will focus on overcoming this limitation by expanding the sample size to include more female athletes and exploring the use of different sensors and data fusion methods.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to participant confidentiality, but are available from the corresponding author on reasonable request.

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References

- Phomsoupha, M. & Laffaye, G. The science of badminton: game characteristics, anthropometry, physiology, visual fitness and biomechanics. *Sports Med.* **45**, 473–495 (2015).
- Chu, W. T. & Situmeang, S. Badminton video analysis based on spatiotemporal and stroke features. In Proceedings of the ACM on international conference on multimedia retrieval, 2017; pp 448–451. (2017).
- Steels, T. et al. Badminton activity recognition using accelerometer data. *Sensors* **20** (17), 4685 (2020).
- Poongodi, T., Krishnamurthi, R., Indrakumari, R., Suresh, P. & Balusamy, B. Wearable devices and IoT. In Balas, V. E.; Solanki, V. K.; Kumar, R.; Ahad, M. A. R. (Eds.) A Handbook of Internet of Things in Biomedical and Cyber Physical System, Springer, 245–273. (2019).
- Kim, K., Jung, M., Jeon, S. & Bae, J. Robust and scalable three-dimensional spacer textile pressure sensor for human motion detection. *Smart Mater. Struct.* **28** (6), 065019 (2019).
- Cheng, Y. et al. Recent developments in sensors for wearable device applications. *Anal. Bioanal. Chem.* **413** (24), 6037–6057 (2021).
- Chander, H. et al. Wearable stretch sensors for human movement monitoring and fall detection in ergonomics. *Int. J. Environ. Res. Public Health*. **17** (10), 3554 (2020).
- Christ, J. F., Aliheidari, N., Pötschke, P. & Ameli, A. Bidirectional and stretchable piezoresistive sensors enabled by multimaterial 3D printing of carbon nanotube/thermoplastic polyurethane nanocomposites. *Polymers* **11** (1), 11 (2018).
- Vu, C. C. & Kim, J. Human motion recognition using SWCNT textile sensor and fuzzy inference system based smart wearable. *Sens. Actuators A: Phys.* **283**, 263–272 (2018).
- Kiang, C. T., Yoong, C. K. & Spowage, A. Local sensor system for badminton smash analysis. In 2009 IEEE Instrumentation and Measurement Technology Conference, ; IEEE: pp 883–888. (2009).
- Pardiwala, D. N., Subbiah, K., Rao, N. & Modi, R. Badminton injuries in elite athletes: A review of epidemiology and biomechanics. *Indian J. Orthop.* **54** (3), 237–245 (2020).
- Lieber, R. L. & Fridén, J. Mechanisms of muscle injury after eccentric contraction. *J. Sci. Med. Sport.* **2** (3), 253–265 (1999).
- Lam, W. K. et al. Understanding the impact loading characteristics of a badminton lunge among badminton players. *PLoS One* **13** (10), e0205800 (2018).
- Vaz, J. R. et al. Muscle coordination during breaststroke swimming: comparison between elite swimmers and beginners. *J. Sports Sci.* **34** (20), 1941–1948 (2016).
- Le Sage, T. et al. Development of a real time system for monitoring of swimming performance. *Procedia Eng.* **2** (2), 2707–2712 (2010).
- Zheng, Y. J. et al. Wearable and wireless performance evaluation system for sports science with an example in badminton. *Sci. Rep.* **12** (1), 16855 (2022).
- Fang, L. & Sun, M. Motion recognition technology of badminton players in sports video images. *Future Generation Comput. Syst.* **124**, 381–389 (2021).
- Van Herbruggen, B. et al. Strategy analysis of badminton players using deep learning from IMU and UWB wearables. *Internet Things*. **27**, 101260 (2024).
- Ghosh, I., Ramamurthy, S. R., Chakma, A., Roy, N. & Decoach Deep learning-based coaching for badminton player assessment. *Pervasive Mob. Comput.* **83**, 101608 (2022).
- Luczak, T., Burch, R., Lewis, E., Chander, H. & Ball, J. State-of-the-art review of athletic wearable technology: what 113 strength and conditioning coaches and athletic trainers from the USA said about technology in sports. *Int. J. Sports Sci. Coaching*. **15** (1), 26–40 (2020).
- Yu, S. et al. Artificial neural network-based activities classification, gait phase estimation, and prediction. *Ann. Biomed. Eng.* **51** (7), 1471–1484 (2023).
- Arlotti, J. S. et al. Benefits of IMU-based wearables in sports medicine: narrative review. *Int. J. Kinesiol. Sports Sci.* **10** (1), 36–43 (2022).
- Hu, G. et al. Model predictive based unscented Kalman filter for hypersonic vehicle navigation with INS/GNSS integration. *IEEE Access*. **8**, 4814–4823 (2019).
- Ishida, K. IoT application in sports to support skill acquisition and improvement. In 2019 IEEE 12th Conference on Service-Oriented Computing and Applications (SOCA), ; IEEE: pp 184–189. (2019).
- Beniczky, S., Karoly, P., Nurse, E., Rylvlin, P. & Cook, M. Machine learning and wearable devices of the future. *Epilepsia* **62**, S116–S124 (2021).
- Wang, Z. et al. Using wearable sensors to capture posture of the human lumbar spine in competitive swimming. *IEEE Trans. Human-Machine Syst.* **49** (2), 194–205 (2019).
- Wang, H., Kang, P., Gao, Q., Jiang, S. & Shull, P. B. A novel PPG-FMG-ACC wristband for hand gesture recognition. *IEEE J. Biomedical Health Inf.* **26** (10), 5097–5108 (2022).
- Kang, P., Li, J., Jiang, S. & Shull, P. B. Reduce system redundancy and optimize sensor disposition for EMG-IMU multimodal fusion human-machine interfaces with XAI. *IEEE Trans. Instrum. Meas.* **72**, 1–9 (2022).
- Franco, T. et al. Motion sensors for knee angle recognition in muscle rehabilitation solutions. *Sensors* **22** (19), 7605 (2022).
- Krizhevsky, A., Sutskever, I. & Hinton, G. E. Imagenet classification with deep convolutional neural networks. Advances in neural information processing systems 2012, 25.
- Li, Z., Liu, F., Yang, W., Peng, S. & Zhou, J. A survey of convolutional neural networks: analysis, applications, and prospects. *IEEE Trans. Neural Networks Learn. Syst.* **33** (12), 6999–7019 (2021).
- Seong, M. et al. Multisensebadminton: wearable sensor-based Biomechanical dataset for evaluation of badminton performance. *Sci. Data*. **11** (1), 343 (2024).
- Qiu, S. et al. Multi-sensor information fusion based on machine learning for real applications in human activity recognition: State-of-the-art and research challenges. *Inform. Fusion*. **80**, 241–265 (2022).
- Lin, J., Chang, C. W., Ik, T. U. & Tseng, Y. C. Sensor-based badminton stroke classification by machine learning methods. In 2020 International Conference on Pervasive Artificial Intelligence (ICPAI), ; IEEE: pp 94–100. (2020).
- Peralta, D. et al. Badminton stroke classification based on accelerometer data: from individual to generalized models. In IEEE International Conference on Big Data (Big Data), 2022; IEEE: pp 5542–5548. (2022).
- Wang, Z., Guo, M. & Zhao, C. Badminton stroke recognition based on body sensor networks. *IEEE Trans. Human-Machine Syst.* **46** (5), 769–775 (2016).
- Srimath, S., Ye, Y., Sarker, K., Sunderraman, R. & Ji, S. Human activity recognition from rgb video streams using 1d-cnns. In 2021 IEEE SmartWorld, ubiquitous intelligence & computing, advanced & trusted computing, scalable computing & communications, internet of people and smart city innovation (SmartWorld/SCALCOM/UIC/ATC/IOP/SCI), ; IEEE: pp 295–302. (2021).
- Zhang, X. Application of human motion recognition utilizing deep learning and smart wearable device in sports. *Int. J. Syst. Assur. Eng. Manage.* **12** (4), 835–843 (2021).
- Fan, J., Bi, S., Wang, G., Zhang, L. & Sun, S. Sensor fusion basketball shooting posture recognition system based on CNN. *J. Sens.* **2021** (1), 6664776 (2021).
- Ma, R. et al. Basketball movements recognition using a wrist wearable inertial measurement unit. In 2018 IEEE 1st International Conference on Micro/Nano Sensors for AI, Healthcare, and Robotics (NSENS), ; IEEE: pp 73–76. (2018).
- Babiuch, M., Foltýnek, P. & Smutný, P. Using the ESP32 microcontroller for data processing. In 2019 20th International Carpathian Control Conference (ICCC), ; IEEE: pp 1–6. (2019).

42. Maier, A., Sharp, A. & Vagapov, Y. Comparative analysis and practical implementation of the ESP32 microcontroller module for the internet of things. In 2017 Internet Technologies and Applications (ITA), ; IEEE: pp 143–148. (2017).
43. Othman, M. A., Amri, L. H., Azmi, M. N. F., Othman, N. A. & Hassan, M. H. A. Wireless Timing System Using ESP-To-ESP Communication Protocol for Precision Timing in Sports. In International Conference on Innovation and Technology in Sports, ; Springer: pp 577–586. (2023).
44. Saavedra, M. B., Merchan, F. T., Astudillo-Salinas, F., Belesaca, J. D. & Barreto, J. Development of an IoT system prototype for the acquisition of biomechanical data of a speed skater. In 2024 IEEE Eighth Ecuador Technical Chapters Meeting (ETCM), ; IEEE: pp 1–6. (2024).
45. Manrique, D. C. & González-Badillo, J. Analysis of the characteristics of competitive badminton. *Br. J. Sports Med.* **37** (1), 62–66 (2003).
46. Macquet, A. C. & Fleurance, P. Naturalistic decision-making in expert badminton players. *Ergonomics* **50** (9), 1433–1450 (2007).
47. Sciascia, A. & Cromwell, R. Kinetic chain rehabilitation: a theoretical framework. *Rehabilitation research and practice* 2012, 2012 (1), 853037.
48. Cust, E. E., Sweeting, A. J., Ball, K. & Robertson, S. Machine and deep learning for sport-specific movement recognition: A systematic review of model development and performance. *J. Sports Sci.* **37** (5), 568–600 (2019).
49. Tabrizi, S. S., Pashazadeh, S. & Javani, V. A deep learning approach for table tennis forehand stroke evaluation system using an IMU sensor. *Comput. Intell. Neurosci.* **2021** (1), 5584756 (2021).
50. Kos, M. & Kramberger, I. Tennis stroke consistency analysis using miniature wearable IMU. In 2018 25th International Conference on Systems, Signals and Image Processing (IWSSIP), ; IEEE: pp 1–4. (2018).
51. Nguyen, L. N. N. et al. Basketball activity recognition using wearable inertial measurement units. In Proceedings of the XVI international conference on Human Computer Interaction, ; pp 1–6. (2015).
52. Fan, J. Z., Liu, X. & Ni, G. X. Angular velocity affects trunk muscle strength and EMG activation during isokinetic axial rotation. *Biomed. Res. Int.* **2014** (1), 623191 (2014).

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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